



UTM
UNIVERSITI TEKNOLOGI MALAYSIA

FACULTY OF COMPUTING
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Design Thinking Project Report

Project Title : Smart Parking on Campus

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1.0 Introduction

1.1 Background Information

According to the Facts and Figures UTM website, the number of students in 2025 is reported to be about 30142 students studying at the campus UTM. For fresh students, personal transport is prohibited on campus since campus buses are provided, but the privilege to use personal transport around campus is held by senior students. Most of the senior students are reported to use transportation like cars and motorcycles as greater convenience and safety are perceived. There is no need to wait for the bus and travel can be done at any time to accommodate schedule changes whatever the class at night. However, the parking space of the campus is harder to find without any new system helping and most impatiently during peak hour or large campus events make the overcrowded, so severe traffic congestion is led to, resulting in significant time wasted in slow-moving or stalled traffic. Therefore, an application is going to be proposed to assist transportation management and park availability on campus, namely “CampusSpot.”

1.2 About Us

“CampusSpot” is defined as a smart parking and traffic management system which means it is designed as an integrated, data-driven mobile and app that uses real-time sensors, predictive analytics and user feedback to optimize parking availability and reduce traffic congestion on campus. In a campus setting, it has been developed with the aid of IoT sensors placed in every parking spot. Further, traffic flow data has been identified by traffic sensors placed around traffic points or existing CCTV cameras and its traffic position by receiving time-stamped signals from multiple satellites. The data generated is then processed on a cloud platform. Through the app, parking spot availability can be accessed at any time of the day and night, and the routing directions for navigation are provided intelligently, taking into consideration the current level of traffic on the road. By significantly reducing time spent searching for parking, minimizing traffic congestion through intelligent routing, a smarter, more efficient, and more sustainable mobility experience is provided to the entire campus community through “CampusSpot”.

1.3 Objective and Overview of This Project

The purpose of this project is to implement a smart parking solution along with a traffic flow management solution with special focus given to accessibility requirements for disabled students in the campus environment. Accessibility improvement is one of the primary goals of this project, focusing on real-time information about accessible parking slots and campus traffic flow. By leveraging the capabilities of IoT along with data analytics, an updated, convenient, and trouble-free campus transport solution is planned to be offered to students and staff through “CampusSpot”. For instance, one can check for the availability of parking space, receive notifications for parking space availability, and use a route direction service for avoiding traffic congestions. Furthermore, a smart navigation system can be incorporated in the project as one of its key components to provide easy access to navigation services in order to avoid congested traffic routes.

2.0 Target Consumer

“CampusSpot” is created to satisfy the constant need for effective, easy, and intelligent options in managing parking on campus. Data from sources from different places is integrated by this solution, and live tracking is added, thus resulting in a much easier parking experience. Mainly, the solution is directed toward college students, faculty, and visitors, and those who use their cars for daily commuting would be the ones benefiting the most from this solution. Consequently, the role of guiding the users to the available spots is played by the system, hence cutting down the search time and reducing the traffic, especially during peak hours. Besides, by giving updates and making suggestions, a part of users’ effective parking planning and decreasing the pressure caused by the lack of parking space is had by “CampusSpot.”

In addition, “CampusSpot” is exactly what busy students attending consecutive classes and employees with rigid schedules who frequently run into the problem of not being able to park on the campus need. It is targeted at the people who go through the same agony in the form of lost time, frustration, and uncertainty caused by packed parking lots. The user's mobility is improved, and proper time management is encouraged by the system with its real-time availability tracking, smart parking recommendations, and notifications features. All in all, “CampusSpot” is a smartly designed and consumer-oriented modern solution that makes daily commuting to and from the campus pleasant through intelligent means.

3.0 Problems and Solutions Problem

Problem	Description	Solution
Difficulty in finding available parking spaces	Daily productivity is frequently diminished, and academic tardiness is increased by the continuous search for parking during peak hours.	Through the implementation of "CampusSpot", real-time status updates are provided, enabling vacant spots to be pre-identified and navigated directly.
Campus traffic congestion	Campus traffic congestion, heightened accident risks, and increased carbon emissions are primarily driven by continuous vehicle circulation.	These environmental and logistical issues are mitigated by "CampusSpot" through the provision of intelligent routing, ensuring that vehicles are directed toward less occupied areas.
Lack of real-time parking information	Traditionally, imprecise guesswork and past practices have been relied on by parking choices, as there was no dynamic and real-time data available.	This problem is solved by "CampusSpot" by providing a single platform where live data is offered, making it possible to make dependable decisions even before arriving at the place.
Poor time management	The time wasted on parking search negatively affects student and staff schedules, making productivity lower and stress levels higher.	With "CampusSpot" in place, commuting will be arranged more efficiently thanks to the predictive alerts, thus keeping the campus appointments more on time.

Table 3.1 Problems and Solutions

4.0 Team Working

Teamwork is critically important for this project. Therefore, to optimize workflow efficiency, tasks were decided to be divided among ourselves by our group. On 20 November 2025, “Smart transportation or parking on campus” was chosen by us as our topic. Day by day, ideas began to be contributed by us on how to implement the topic smoothly.

Google Form data collection was created and distributed by Najla. Additionally, real-time surveys were conducted by all members through interviews with students and lecturers, with the interviews being led by Li Ning. After that, the prototype was developed by us. The report was also contributed to by all group members, and presentation slides were created. Video editing was focused on by Nurain.

Throughout this design thinking process, we continuously supported each other to ensure that everything progressed well. All collaborations were conducted online via our WhatsApp group.

5.0 Design Thinking

5.1 Design Thinking Process

5.1.1 Empathy

The first stage of the design thinking process, empathy, focuses on user-centric research. To gain a profound comprehension of the issue that we intend to resolve is the objective. The main aim of the empathy stage is to uncover the user's needs and the hidden difficulties that have a say in the progress of the product we wish to develop.

Up to this time, user observations have been conducted by us to deeply understand the users' environment and generate empathy. For this purpose, a Google Form was developed by us to gather data from the community, focusing on students who frequently drive to school. Data on their everyday parking challenges, how long they spend looking for parking, trouble times, and proposals for improvements was collected by us through the survey. Furthermore, a brief interview was done by our team with some UTM students and one UTM lecturer to better understand personal experiences, pains, and hopes regarding parking on campus.

Based on the survey responses and interview findings, several recurring issues faced by users were identified by us. These include difficulty in finding available parking spaces during peak hours, lack of real-time parking information, increased stress due to uncertainty, and time wastage caused by repeatedly circling parking areas. These insights helped our group better understand user pain points and guided the development of the “CampusSpot” prototype.

5.1.2 Define

During the Define stage, the data gathered from the Empathy stage, such as the results of the survey and the feedback gathered from the interviews, was organized and analyzed by our group. Our observations are analyzed by us to identify and articulate the core problems faced by users. After collecting the data and conducting interviews, a discussion is engaged by our team to review results and determine the problems faced by the user, thereby developing the problem statement.

5.1.3 Ideate

The Ideate stage was served as the critical transition from understanding campus parking frustrations to engineering the “CampusSpot” solution. Our process began with a "divergent thinking" phase, where the quantity of ideas was prioritized over their immediate perfection by the team. "How Might We" inquiries derived from previous user insights like the anxiety of being late for class were adopted, and confirmation was given that the whole range of proposed features, including the real-time displays and predictive alerts, was established on an authentic user demand. This not only made the brainstorming process more focused and less random but also revealed the specific user vision that could be very well served with real solutions.

As part of the brainstorming process, a shift was made towards a "convergent" phase characterized by the justification and picking of the solutions that appeared most effective. Each idea was subjected to analysis through an Impact vs. Feasibility matrix, which granted the opportunity to consider the technical challenges in proportion to the probability of time-saving and better decision-making. Specific justifications were provided for the inclusion of real-time availability displays and smart recommendations on the basis that they do connect directly to the root cause the lack of actionable data during peak times. Unlike a simple map, a proactive experience is provided by these features, allowing alternative lots to be pivoted to by users before they even enter a congested area. This logical filtering process transformed a wide range of opinions into the focused, user-friendly, and technically viable framework that is now defining the “CampusSpot” prototype.

5.1.4 Prototype

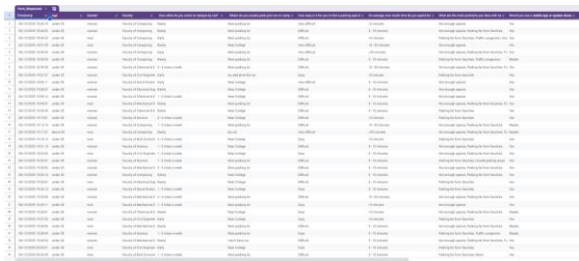
The next stage is the Prototype phase, where the focus is placed on identifying the best possible solutions to each of the problems identified in the earlier three phases. After a series of discussions, it was decided that building a new application named “CampusSpot” could provide a better user experience and greater convenience than already existing smart parking kiosk applications. During this phase, ideas and concepts are made tangible. After consultations about the design and functionalities of the application, the design prototype was created using the Canva application, as it was considered preferable when it comes to design.

5.1.5 Test

The last stage is the testing phase, which is an essential step to ensure that the “CampusSpot” system runs efficiently. The testing on the system will be done to be able to address any possible problems in a better way. The tutorial video, including all the operating steps, was created for users to easily follow and give relevant feedback regarding it. The “CampusSpot” prototype was also shown to students in faculty mechanical. Most were satisfied with the solution, especially features like accessible parking navigation and streamlining traffic with smart routing.

5.2 Design Thinking Evidence

5.2.1 Sample Work

	Data Collection The problems faced by users and suggestions from them were collected by us.
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Interview session

Interviewer: Lim Li Ning

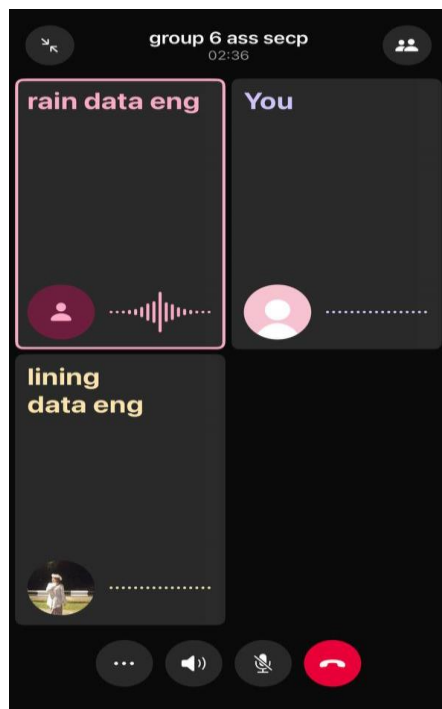
Interviewee: Ahmad

The challenges he experienced while finding parking spaces on campus and his suggested solutions for improvement were asked about by us.



Discussion

The topic of smart parking on campus was chosen by us.



Further Discussion

The problems faced by users were analyzed by us and all possible solutions were listed out by us. Then the best one was chosen by us after a comparison was made.

	Building a prototype After the discussion, a prototype started to be built by us.
	Testing When our prototype was done, our prototype was demonstrated and tested with different users by us.

Table 5.2.1 Design thinking evidence

5.2.2 Record for each phase

1. Empathy

For the Empathy phase, an interview session was conducted by us with two respondents who regularly face parking challenges in their daily lives. Below is the list of questions asked during the interview:

- On a scale of 1-10, how stressful is finding parking on campus during peak hours, and what specifically makes it stressful? (Respondent #1)

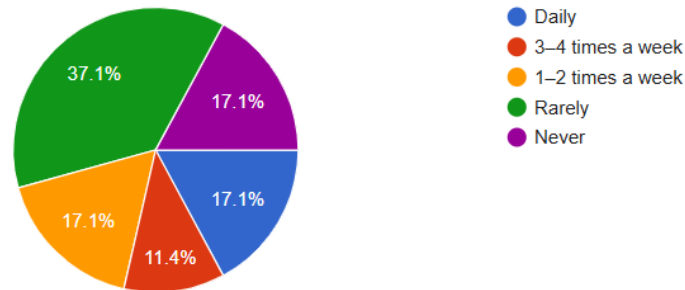


- Would you use a mobile app that shows real-time parking availability? Would you pay for premium features? (Respondent #2)



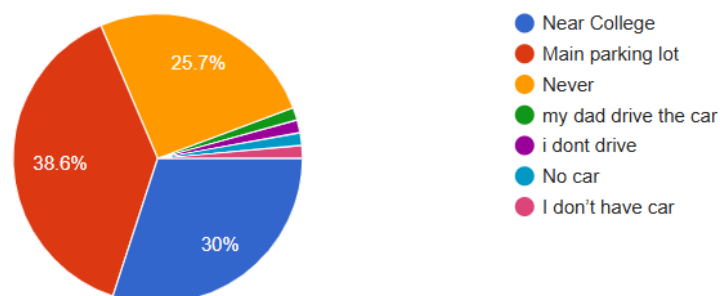
A survey was also generated by us using Google Forms to get more information from a wider range of people gathered from a diverse sample of 70 respondents across various academic faculties. Below is the result from Google Forms:

How often do you come to campus by car?
70 responses



The frequency with which students commute to class by car is illustrated by the first pie chart, revealing that a vast majority are daily drivers who rely on personal transport to navigate their academic schedules.

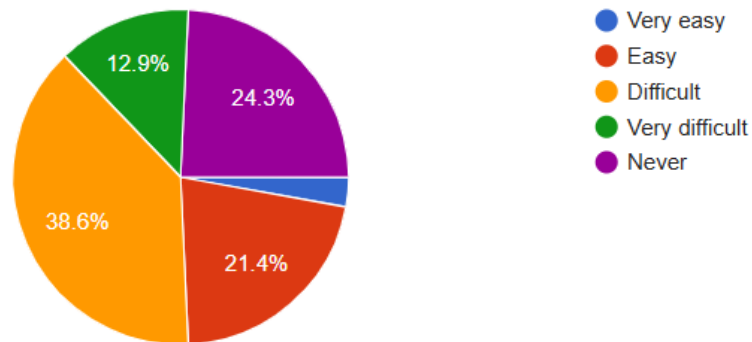
Where do you usually park your car on campus?
70 responses



The parking patterns of the users are presented by the second pie chart, with a clear preference of the users parking in lots adjacent to the faculty buildings. A very critical bottleneck is provided by such data, since the spots that are closest to their classrooms are chosen by the students, and thus very extreme competition exists for these high-demand zones with limited availability.

How easy is it for you to find a parking spot during peak hours?

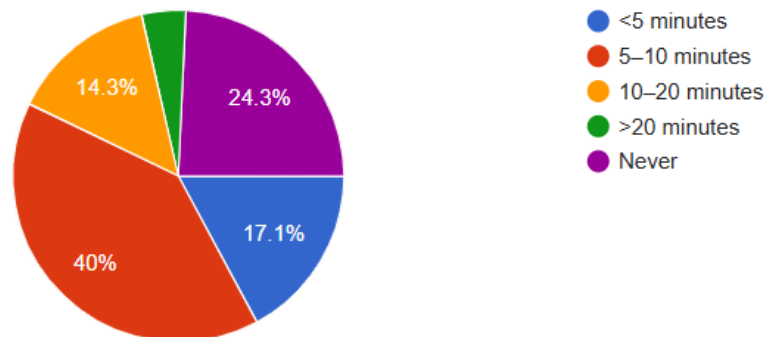
70 responses



The logistical challenges the existing parking system faces through the view of the users' ease of getting a spot during peak hours are illustrated by the third pie chart. The process is "difficult" by a large majority of those who answered the question, which is a significant area of concern from the user's perspective.

On average, how much time do you spend looking for a parking spot?

70 responses

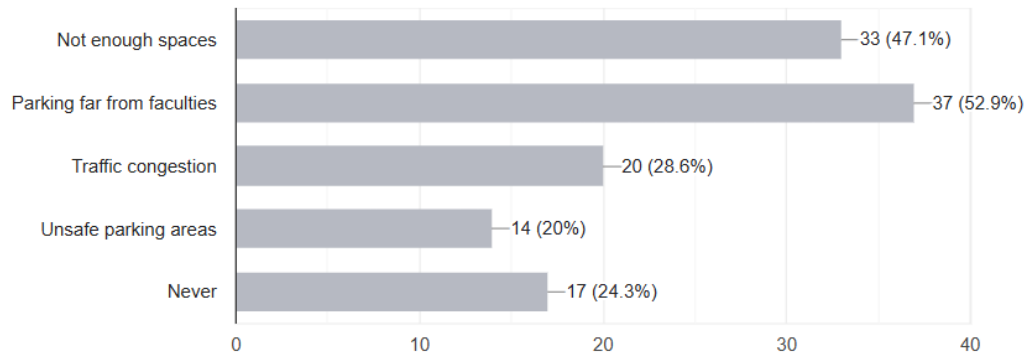


The frequency of parking searches that take a certain period is presented by the fourth pie chart, indicating that 5 to 10 minutes are being spent by a major part of the student body waiting for a vacancy. Solid proof of continuous wasting and inefficiency is given by all these images, causing the users' annoyance, leading to overloading and tardiness in academics.

What are the main problems you face with campus parking? (can choose more than one)

 Copy chart

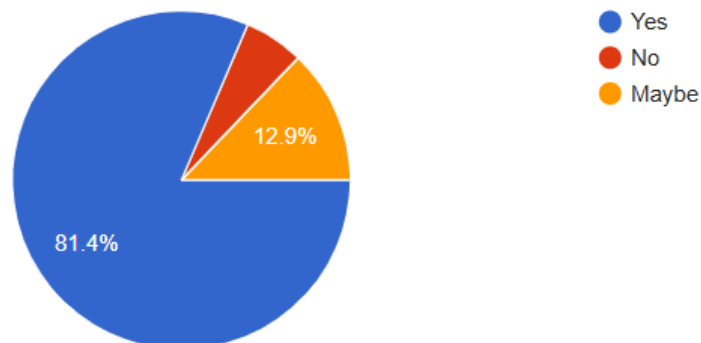
70 responses



By using a bar chart, parking distances from faculties, not enough spaces, bad traffic congestion, and safety hazards due to darkness in out-of-way parking lots are reported as the main issues.

Would you use a **mobile app or system showing real-time parking availability**?

70 responses



User receptivity toward a digital intervention is measured by the last pie chart. A clear mandate for the project is provided by the overwhelming agreement among respondents for a mobile application.

2. Define

The table below shows some problems that users face.

PROBLEMS	DESCRIPTIONS
Unsafe parking areas	The insufficient lighting in the outer parking areas is made to be easier for criminals to commit acts like vandalism, which in turn discourages the area from being used at night by the campus community. For the sake of user trust and space occupancy being optimized, security measures should be integrated, and lighting should be improved to create a safer parking area.
Traffic congestion	The lack of real-time traffic information has negative repercussions in the form of a traffic jam that is so bad and long wait times at the same time during busy hours, which, of course, driver patience decreases greatly. A system that automatically manages and gives live directional guidance is a must to eliminate delays and make the campus traffic flow smoother.
Parking far from faculties	The long distance separating the remote parking lots from the faculty buildings is a major obstacle for users, especially if it is bad weather or they are physically challenged. A wastage of parking spaces is caused by the distance; therefore, it is better to offer nearer parking places or to connect the campus with more transit links to make it accessible all over.
Not enough spaces	The peak time in parking demand is often higher than the current infrastructure capacity, which causes loss of productive time and unauthorized parking due to the absence of occupancy data. Real-time availability without a central communication platform still makes parking management ineffective and thus continues to aggravate the users' stress levels.

Table 5.2.2.1 Problems and Description

3. Ideate

The table below shows the possible solutions that can be used to solve users' problems.

PROBLEMS	SMART KIOSK SYSTEM	CAMPUSSPOT APP
Unsafe parking areas	The users will be informed of lit and monitored parking areas by the smart kiosk on a digital screen for them to make a safer choice.	Parking spaces that are secure, well-lit, and under surveillance are marked by the application. Places that are less risky in terms of safety are assisted in being selected by the drivers.
Traffic congestion	Available parking areas near faculties are displayed by the smart kiosk and users are guided with digital signage.	Parking spaces that are secure, well-lit, and under surveillance are marked by the application. Places that are less risky in terms of safety are assisted in being selected by the drivers.
Parking far from faculties	Available parking areas near faculties are displayed by the smart kiosk, and users are guided by it with digital signage.	Parking areas near faculties that are vacant are displayed by the application, and parking spots are let choose by users for minimizing walking distance.
Not enough spaces	A vacant parking area is located by the smart kiosk, and the individual is guided to that specific area by it.	Real-time available parking spaces are displayed by the application. Areas with vacant spots can be quickly identified by users.

Table 5.2.2.2 Possible solutions to solve the problems

As shown in Table 5.2.2.2, the development of a "CampusSpot" application was selected as the most feasible solution due to its potential to enhance accessibility, provide real-time information, and improve campus mobility. The conceptual design will be transformed into a tangible system in the Prototype phase through further development of this solution.

4. Prototype

After our discussion, it was concluded by us that developing a new application would provide a better user experience and better convenience.

The reasons for the decision are showed in the table below:

REASONS	SMART KIOSK SYSTEM	CAMPUSSPOT APP
Cost-effectiveness	Need higher cost due to hardware, installation and ongoing physical maintenance.	Lower cost as only required software
Accessibility	Required finding specific kiosk spots on campus	Can be used anytime through smartphone so that no need to visit fixed locations in campus
Coverage	Limited area that near kiosk locations	Campus-wide, supports both indoor and outdoor navigation
Update And Maintenance	Required on-site hardware maintenance and updates	Real-time updates through cloud, no physical maintenance needed
User Engagement And Interactivity	Medium engagement and interactivity since touch only interaction, no personalization	High engagement and interactivity since including touch, voice notifications and personalization.

Table 5.2.2.3 Comparison between Smart Kiosk System and Application

Since the comparison shown in table 5.2.2.3, the decision was made to develop the “CampusSpot” application that allows users to see which parking spots are free in real time and suggests the best routes to avoid traffic. A Smart Parking Broadcast Assistant that gives user personalized parking updates during the period that you choose is also included.

The table below shows the features of our application:

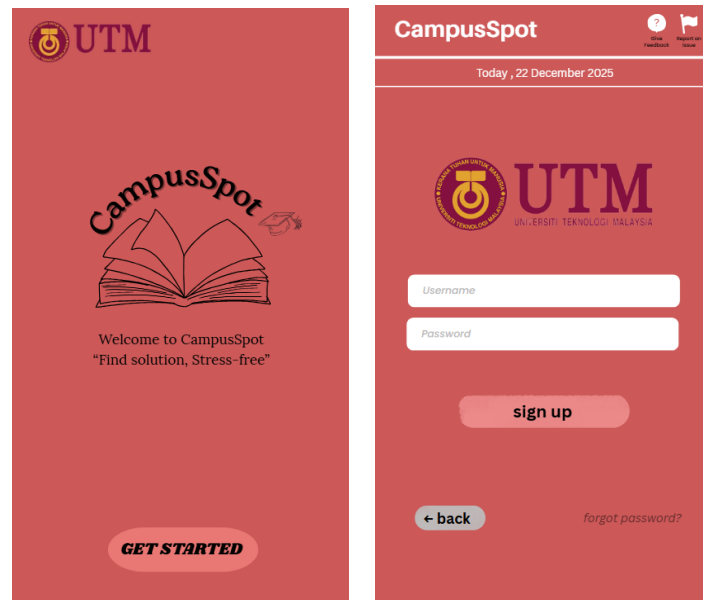
FEATURES	DESCRIPTION
Real-time Parking Status Update	Live availability of parking spaces across campus is displayed by “CampusSpot” using color-coded mapping which is green to represent available and red to represent fully.
Parking Reminders	A specific location can be set, and a notification interval can be selected to get notifications so that notifications can be received automatically before closing the notification by the user.
Traffic Flow View	A live traffic map made specifically for campus is provided by "CampusSpot." Busy or clear roads are shown based on your search, and better routes are suggested so that users aren't delayed by traffic.
Smart Navigation	Available traffic roads are guided to users by “CampusSpot,” as current traffic flow is analyzed to ensure an efficient arrival at their destination.

Table 5.2.2.4 Features for “CampusSpot” application

The improvement of campus mobility, the reduction of congestion, and the enhancement of accessibility for disabled students are shown in Table 5.2.2.4. In the Prototype phase, the development of the selected solution is ensured to transform conceptual designs into a tangible system.

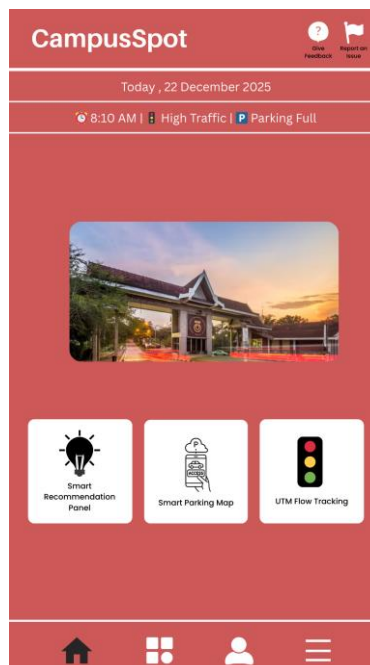
Prototype Showcase

1.



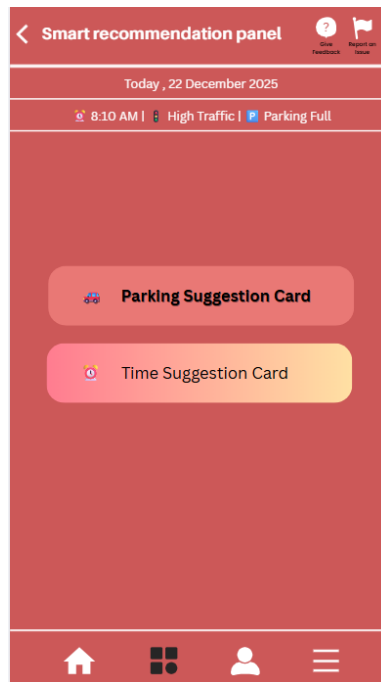
A clean design that ensures security is provided with a sign-up interface for staff or student UTM accounts (new user).

2.



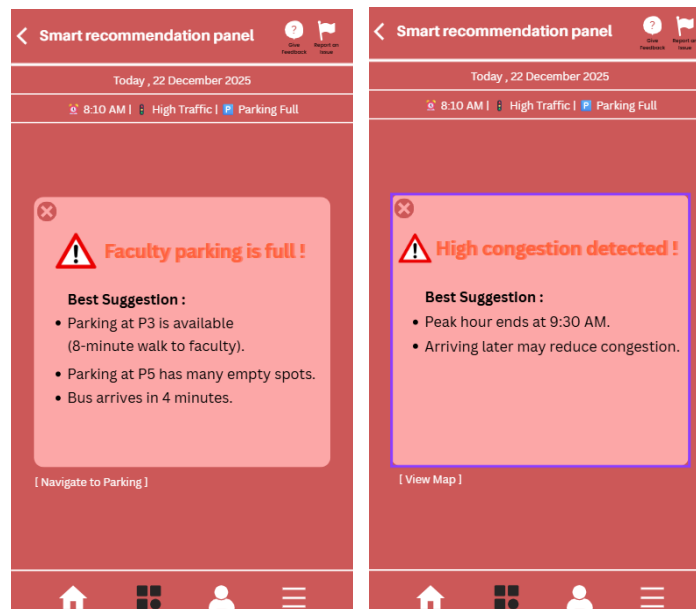
Three options are offered for choosing the panel, which are Smart Parking Map, UTM Flow Tracking, and Smart Recommendation.

3.



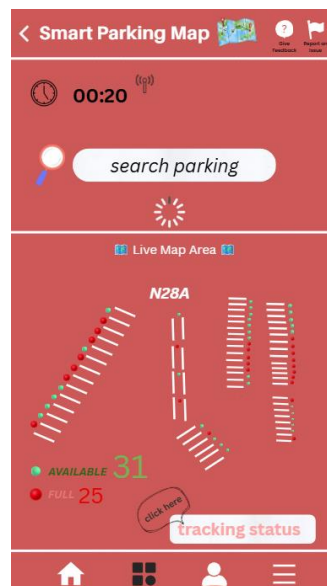
Smart Recommendation is shown by the prototype, and two options are included, Parking Suggestion Card and Time Suggestion Card.

4.



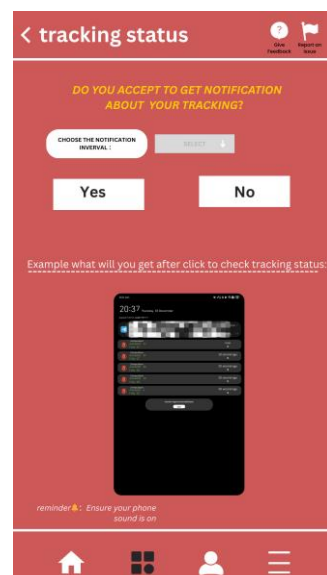
Some notices are contained in the Parking Suggestion Card and the Suggestion Card to advise the user.

5.



Available parking spots in real time are shown by the Smart Parking Map, so a space can be found quickly. The time is always auto-updated by the timer. An example of the live map area is shown in the prototype. The parking space layout, available parking, and full parking are shown in the Live Map Area. A button labeled "tracking the status" is placed at the bottom and can be accessed with just one tap.

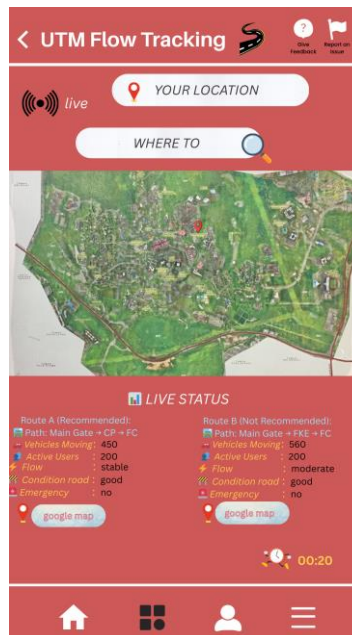
6.



The tracking status is shown by the prototype when the previous button is tapped. A notification of interval should be set by the user for getting live updates periodically. When the "yes" button is tapped, notifications will be received from the system. The user is

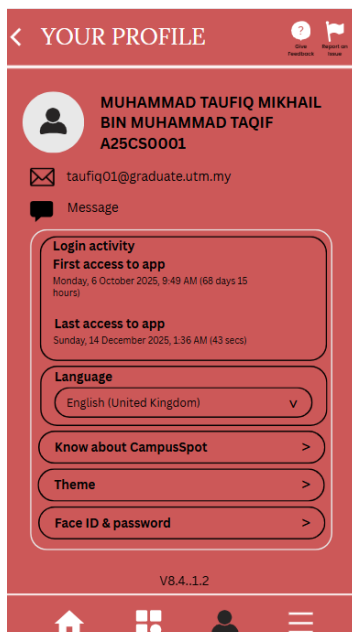
reminded to keep the phone sound on. When "no" is clicked, the Smart Parking Map is shown again.

7.



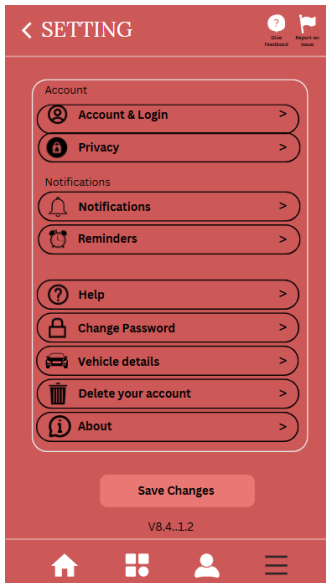
UTM Flow Tracking is shown by the prototype. The location should be pinpointed by the user by opening their location or entering it, and then the destination should be entered. The map, the location to go, and the road condition will be shown by the live status. All routes to the destination are supported by vehicles moving, active users, flow, road condition, and emergency. Google Map can be clicked by the user to get the route.

8.



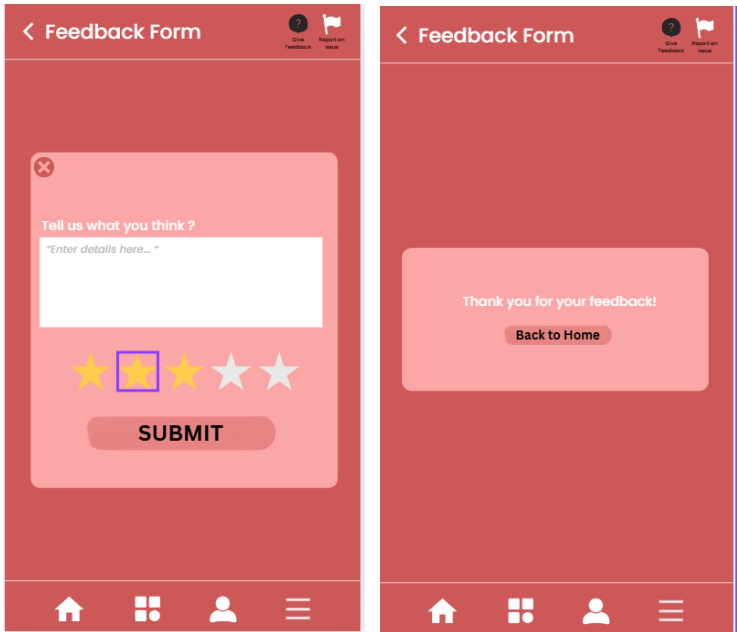
The user profile is shown by the prototype. Activity of the user can be checked and the language like English, Chinese, and Malay can be chosen. Know about “CampusSpot” can be clicked to obtain more information. For security purposes, FaceID and password authentication can be set up. The theme can also be changed based on the user’s preferences and comfort.

9.



Management of the account, notifications, vehicle info, and preferences is allowed by the Settings center in one place.

10.



The feedback form and report are shown by the prototype. Suggestions and specific issues can be entered by the user. Then, the feedback will be received, and improvements will be made by our platform after submit button is clicked.

Through 10 prototype showcases, it is shown that an all-in-one campus mobile app is combined with real-time parking navigation, route suggestions, personalized settings, and feedback forms to make the user more effective and stress-free.

5.Test

Once the “CampusSpot” prototype was completed, it was presented to Miss Soffia from the Faculty of Mechanical Engineering, who is a second-year senior, and its practical design and functionality were commended by her. In recognition of the feedback from Miss Soffia, the intuitive design was specifically praised, and it was noted that real-time parking tracking and data-driven route suggestions are effectively targeted at the problems faced by students, such as traffic congestion and unavailable parking slots.

5.3 Design Thinking Assessment

5.3.1 During the end of the project demonstration

To conclude our project demonstration, an app called “CampusSpot” has been created by our group. The “CampusSpot” application was developed to improve parking space usage around the campus. To understand the issues faced by users, feedback is being gathered through surveys and interviews with students and staff. This feedback is helping us to understand the problems related to parking. Based on this data, ideas have been generated and features have been designed by our group, aimed at making parking more efficient and convenient. Team discussions were held to ensure agreement on the final concept and ensure the app’s design worked effectively. The main features of the application were presented in the final demonstration, showing how actual parking issues on campus could be solved.

5.3.2 During the transition between design thinking phases

In the transition between phases of the design thinking process, during the Empathize phases, data about the parking space issues faced by the students and staff was collected via Google Form and interview. The problems are analyzed and the solution that meets everyone's needs is found before moving to the ideation phases, where potential solutions are discussed and brainstormed. During the discussion, it is ensured that each idea that is agreed upon will address the identified issues effectively. After the best solution is selected, a prototype is built, and feedback is gathered through early testing that informed improvement before moving to the next phases. This ongoing assessment between phases allowed problems to be identified early and ensured that “CampusSpot” meets the needs of the users.

6.0 Reflections

1. What is your goal/dream with regard to your course/program?
2. How does this design thinking impact on your goal/dream with regard to your program?
3. What is the action/improvement/plan necessary for you to improve your potential in the industry?

NAME	REFLECTION
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LIM LI NING	1. The goal of becoming a professional data engineer is pursued. Since the course is focused on data engineering, more struggles should be dedicated to designing and building robust data, managing infrastructure, and ensuring quality; more essentially, collaboration with the team is required. Not only that, all basic skills and soft skills are being learned to feel more confident in preparing for a future career in computer science. 2. Communication skills between different ethnicities were enhanced by this project. Before this project, a generation gap with teammates was felt. However, as problems were solved together and ideas were contributed to finishing the project, communication between ethnicities was made much easier. Besides that, writing skills were learned through the project report. Although writing is not considered a strongest suit, a chance to practice it was given by working on this report. For instance, the details of IoT were forced to be learned using design thinking. It is believed that these skills will be helpful in a future career and in the next project and assignment. 3. To enhance potential in the industry, more reading is planned to be done to generate more ideas about design thinking, and deeper insights will be gained through YouTube, news, and other platforms. Concepts will be focused on and continuous learning will be maintained in the future or currently. Additionally, more projects will be done by joining a club, such as Airost, to ensure more is learned.
NAJLA AUNI BINTI MOHAMAD ASRI	1. Working with data has always been my dream, and the Data Engineering course is the initial step I take to gain a strong foundation in data management, programming, and analytics. My aim is to master the skills required for handling large datasets in terms of gathering, processing, and analyzing them quickly and easily. I aspire to be an effective

	and professional data engineer who is involved in the production of data pipelines and systems that empower industries with the power of data-driven decision-making. 2. Design thinking is a methodology that helps me to see challenges in a creative way as well as in a structured way. I can achieve that through the user empathy, problem definition, solution brainstorming, prototyping, and testing process to find the least complex and most user-friendly data solutions. With this mindset, I not only limit myself to coding or data manipulation but also come to understand how the data solutions affect real-world decisions; this is in line with my aim of being a powerful data engineer. 3. To make myself more valuable in the field of data engineering, it will be made a point by me to practice programming languages, particularly Python and SQL, on a regular basis. Besides, small personal projects will be worked on by me to acquire practical skills in data collection, cleaning and analysis, and workshops, competitions, or internships will be participated in by me for the sake of gaining industry exposure. Not only that, but design thinking will also be applied by me all the time to come up with creative ways to solve problems and develop solutions that are both efficient and user-focused.
NURAIN NABILAH BINTI AZMAN	1. Enrolling in Computer Science is one of the biggest steps taken by me as I have never been involved in any Computer Science related course in my school, so a strong foundation in data management, programming, and solving problems related to Computer is my goal. Besides that, my soft and hard skills are to be improved by me because Data Engineer is the course that was focused on by me, I hope to be a professional data engineer who can design, create, and manage reliable data systems. The chance to work in a job

	related to data engineering is eagerly looked forward to by me. 2. In my opinion, my soft and hard skills have been improved by the Design Thinking project. Through this design thinking, important soft skills such as communication, teamwork, time management, problem-solving, and critical thinking have been developed by me. Furthermore, a deep understanding of how to write a proper report really helped to be developed by me while working with my teammates. A lot of experience such as creating a prototype, editing a video presentation, and confidence to embrace myself to share my opinions and ideas in a team have also been gained by me. 3. In the future, club and community activities are planned to be actively involved in by me to enhance my soft skills. Time management is also aimed at improving and leadership skills are aimed at being developed by me by taking on responsibilities and working with others. Next, programs related to technology are planned to be participated in by me to gain more exposure to technology such as IT, IoT and AI to make myself stand out for my resume in the future. Strong connections with the company of my dreams are also aimed at being built by me by joining or participating in programs provided by the company such as industry talk, mini projects for students and so on.
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7.0 References

UTM Smart Transportation Survey. (n.d.). Google Docs.
<https://docs.google.com/forms/d/e/1FAIpQLSc9413fogHBzAm8hbIJFRvE1pYukO9hUmqUqUCU4vrzMysAA/viewform>

Video Link : <https://youtu.be/L8OORM-LJPc?si=i3AjkzT-zcsuvs>